Part C

In this part of the test, there are two texts about different aspects of health care. For **questions 1 to 8**, choose the answer (**A**, **B**, **C** or **D**) which you think fits best according to the text.

Write your answers on the separate **Answer Sheet**.

Detecting Carbon Monoxide Poisoning

Carbon Monoxide (CO) poisoning is the single most common source of poisoning injury treated in US hospital emergency departments. While its presentation is not uncommon, the diverse symptoms that manifest themselves do not lead most clinicians to consider carboxyhemoglobinemia when attempting a diagnosis. The symptoms can be mistaken for those of many other illnesses including food poisoning, influenza, migraine headache, or substance abuse.

What's more, in an attempt to find the causative agent for the symptoms, many unnecessary, and sometimes resource-intensive, diagnostics may be ordered, to no avail. For example, because the symptoms of CO poisoning may mimic an intracranial bleed, the time needed to obtain a negative result may hold up a proper diagnosis as well as needlessly increasing healthcare costs. Of even greater concern, however, is that during such delays patients may find that their symptoms abate and their health improves as the hidden culprit, CO, is flushed from the blood during the normal ventilation patterns.

Indeed, multiple reports have shown patients being discharged and returned to the very environment where exposure to CO took place. Take the case of a 67-year-old man who sought medical help after three days of lightheadedness, vertigo, stabbing chest pain, cough, chills and headache. He was admitted, evaluated and discharged with a diagnosis of viral syndrome. Ten days later, he returned to the Emergency Department with vertigo, palpitations and nausea but was sent home for outpatient follow-up. Four days later, he presented again with diarrhea and severe chest pain, collapsing to the floor. This time, he was admitted to the Coronary Care Unit with acute myocardial infarction. Among the results of a routine arterial blood gas analysis there, it was found that his carboxyhemoglobin (COHb) levels were 15.6%. A COHb level then obtained on his wife was 18.1%. A rusted furnace was found to be the source.

There are two main types of CO poisoning: acute, which is caused by brief exposure to a high level of carbon monoxide, and chronic or subacute, which results from long exposure to a low level of CO. Patients with acute CO poisoning are more likely to present with more serious symptoms, such as cardiopulmonary problems, confusion, syncope, coma, and seizure. Chronic poisoning is generally associated with the less severe symptoms. Low-level exposure can exacerbate angina and chronic obstructive pulmonary disease, and patients with coronary artery disease are at risk for ischemia and myocardial infarction even at low levels of CO.

Patients that present with low COHb levels correlate well with mild symptoms of CO poisoning, as do cases that register levels of 50-70%, which are generally fatal. However, intermediate levels show little correlation with symptoms or with prognosis. One thing that is certain about COHb levels is that smokers present with higher levels than do non-smokers. The COHb level in non-smokers is approximately one to two percent. In patients who smoke, a baseline level of nearly five percent is considered normal, although it can be as high as 13 percent. Although COHb concentrations between 11 percent and 30 percent can produce symptoms, it is important to consider the patient's smoking status.

Regardless of the method of detection used in emergency department care, several other variables make assessing the severity of the CO poisoning difficult. The length of time since CO exposure is **one such factor**. The half-life of CO is four to six hours when the patient is breathing room air, and 40-60 minutes when the patient is breathing 100 percent oxygen. If a patient is given oxygen during their transport to the emergency department, it will be difficult to know when the COHb level hit its highest point. In addition, COHb levels may not fully correlate with the clinical condition of CO-poisoned patients because the COHb level in the blood is not an absolute index of compromised oxygen delivery at the tissue level. Furthermore, levels may not match up to specific signs and symptoms: patients with moderate levels will not necessarily appear sicker than patients with lower levels.

In hospitals, the most common means of measuring CO exposure has traditionally been through the use of a laboratory CO-Oximeter. A blood sample, under a physician order, is drawn from either venous or arterial vessel and injected into the device. Using a method called spectrophotometric blood gas analysis, this then measures the invasive blood sample. Because the CO-Oximeter can only yield a single, discrete reading for each aliquot of blood sampled, the reported value is a non-continuous snapshot of the patient's condition at the particular moment that the sample was collected. It does, however, represent a step in the right direction. One study found that in hospitals lacking such a device, the average time

it took to receive results of a blood sample sent to another facility was over fifteen hours, compared to a ten-minute turnaround in CO-Oximeter equipped hospitals.

Text 1: Questions 7 to 14

1 In the first paragraph, what reason for the misdiagnosis of CO poisoning is highlighted?

A the limited experience physicians have of it

B the wide variety of symptoms associated with it

C the relative infrequency with which it is presented

D the way it is concealed by pre-existing conditions

2 In the second paragraph, the writer stresses the danger of delays in diagnosis leading to

A the inefficient use of scarce resources.

B certain symptoms being misinterpreted.

C a deterioration in the patient's condition.

D the evidence of poisoning disappearing.

3 The 67-year-old man's CO poisoning was only successfully diagnosed as a result of

A attending an outpatient clinic.

B his wife being similarly affected.

C undergoing tests as an inpatient.

D his suggesting the probable cause.

4 In the fourth paragraph, confusion is given as a symptom of

A short-term exposure to high levels of CO.

B repeated exposure to varying levels of CO.

C a relatively low overall level of exposure to CO.

D sustained exposure to CO over an extended period.

5 In the fifth paragraph, what point is made about COHb levels?

A They fail to detect CO poisoning in habitual smokers.

B They are a generally reliable indicator of CO poisoning.

C They correlate very well with extreme levels of CO poisoning.

D They are most useful in determining intermediate levels of CO poisoning.

6 The phrase **'one such factor'** in the sixth paragraph refers to

A a type of care.

B a cause of difficulty.

C a method of detection.

D a way of making an assessment.

7 One result of administering oxygen to CO poisoned patients in transit is that

A it becomes harder to ascertain when the COHb level peaked.

B it may lead to changes in the type of symptoms observed.

C it could artificially inflate the COHb level in the short term.

D it affects the ability to assess the effects at tissue level.

8 What reservation about the CO-Oximeter does the writer express?

A It does not always give an immediate result.

B Its use needs to be approved by a physician.

C It requires a skilled analyst to interpret the readings.

D It does not show variations in the patient's condition.